Each criterion was evaluated against every other using a pairwise comparison method. Values were assigned in 0.25 increments between 0 and 1, where:

• A value of 0.5 indicates equal importance.
• Values of 0.25 or 0.75 indicate one criterion is moderately more important than the other.
• Values of 0 or 1 reflect extreme preference toward one criterion, rendering the other negligible.

The comparisons are symmetric such that the sum of each pair equals 1. Total weights were summed and normalized to create priority values for each criterion.

Criteria	Load Bearing Capacity	Thermal Conductivity	Mass per leg	Safety Margin at Yield	Manufacturability & TRL	Cryogenic Ductility	Weight	Normalized
Load Bearing Capability		0.75	0.75	0.5	0.5	2.5	0.25	0.07
Thermal Conductivity	0.25		0.5	0.5	0.5	1.75	0.18	0.05
Mass per Leg	0.25	0.5		0.5	0.5	1.75	0.18	0.05
Safety Margin at yield	0.5	0.5	0.5		0.5	2.5	0.25	0.07
Manufacturing & TRL	0.5	0.5	0.5	0.5		0.5	2.5	0.74
Cryogenic Ductility	0.5	0.5	0.5	0.5	0.5		2.5	0.74
						Total	3.36	

	Load Bearing (12g/6g)	Mass per Leg	Safety Margin at Yield	Thermal Conductivity	Manufacturability & TRL	Cryogenic Ductility
1	Fails load envelope; not viable	>4.0 kg	SF <1.0; failure imminent	>2.5 W/K (unacceptable heat leak)	TRL 4; concept-only; significant development risk	Material unsuitable for cryogenic operation
2	Marginal; requires major redesign	3.25–4.0 kg	SF <1.2; high risk	1.5–2.5 W/K (high; significant thermal penalty)	TRL 5; new design approach; prototype required	Charpy <3 J/cm² at 90 K; high fracture risk
3	Handles 12g/6g with 65–80% yield; major FEA iteration	2.75–3.25 kg (acceptable with tradeoffs)	SF 1.2–1.5	1.0–1.5 W/K (acceptable; manageable heater load)	TRL 6; custom design; requires specialized welding or forming	Charpy 3–5 J/cm² at 90 K (marginal; risk of brittle fracture)
4	Handles 12g/6g with 50–65% yield; minor FEA refinement	2.3–2.75 kg (target range)	SF 1.5–1.8	0.5–1.0 W/K (good; minor heater overhead)	TRL 7; minor customization; standard shop processes	Charpy 5–8 J/cm² at 90 K (acceptable; verified by test)
5	Handles 12g/6g with <50% yield; validated by FEA	<2.3 kg	SF > 1.8	<0.5 W/K (excellent passive isolation)	TRL 8+; off-the-shelf or heritage design; simple machining	Charpy >8 J/cm² at 90 K; excellent toughness

Decision Criteria		Analysis of Landing structure					
		Baseline: Tapered Strut Al 7075-T73		Cross-Braced Tube + Diagonals Ti-6Al-4V		Folding Hinge Leg (Hybrid)	
Criteria	Weight	Score	Weighted Score	Score	Weighted Score	Score	Weighted Score
Load Bearing Capability (12g vert / 6g lateral)	0.25	5	1.25	5	1.25	4	1.00
Mass per Leg (target <2.75	0.20	4	0.80	3	0.60	3	0.60
Safety Margin at yield (target SF > 1.5)	0.25	5	1.25	5	1.25	4	1.00
Thermal COnductivity (minimize deck hear leak at 90K	0.18	3	0.54	2	0.36	4	0.72
Manufacturing & TRL	0.25	5	1.25	3	0.75	2	0.50
Cryogenic Ductility (Charpy >5 J/cm^2)	0.05	4	0.20	5	0.25	3	0.15
Total Score		5.29		4.46		3.97	